# إسئلة سابقة ..

# اقتصاد هندسي


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Polytechnic

**Al-Balqa Applied University**
**Engineering Economy final Exam**

**Student Name.....** **Time: 100 min.**

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**Use the given data in the following Table to solve Q1 to Q3**

| Items | Alt.A | Alt.B | Alt.C | Alt.D |
|---|---|---|---|---|
| F.C | 140,000 | - | 15,000 | 65,000 |
| Extra cost @end of 3rd yr | 20,000 | - | 4,000 | - |
| AOC | - | - | 1,000 | 2,500 |
| A.C for the first 5yrs | 5,500 | - | | |
| Cost every 6 months | - | 500 | | |
| Extra cost every 5 yrs | 8,000 | - | | |
| A.I for the first 5yrswith decreasing gradient(G=400) | - | - | 3,000 | - |
| S.V | - | - | 7,000 | 5,000 |
| Life(n) yrs | 8 | 7 | 5 | 4 |
| i% | 8% | 20%per yr compounded quarterly | 8% | MARR=10% |

**Q1:** Using the capitalized cost criterion which is the optimal alternative A or C? (15 marks)

a- CC1(A) is
b- CC2(A) is
c- CC3(A) is
d- CC4(A) is

e- CC (A) is .....

f- CC1(C) is
g- CC2(C) is
h- CC3(C) is
i- CC4(C) is
j- CC5(C) is
k- CC(C) is .....

**Q2:** The future worth for alternative B is.....

**Q3: Is alternative D**
**1-The value of ($i^*$) is:.....**
**2- Is alternative D Acceptable or not ..** **(10 marks)**

**Q4: If JD 2000 is deposited each semiannual for 5 years at an interest rate of 10 % yearly compounded semiannually , the future worth is.....** **(10 marks)**

**Q5: The total amount due after 7 years for a loan of 30000 at 20% simple interest rate per yea is.....** **(5 marks)**

Al-Balqa Applied University
Faculty of Engineering Technology
Civil Engineering Department
Engineering Economy final Exam 2010/2011

Lecture time Student Name.. Time: 2H. (B)

Use the given data in the following Table to solve Q1 to Q3

| Items | Alt.A | Alt.B | Alt.C | Alt.D | Alt.E |
|---|---|---|---|---|---|
| F.C | 8,000 | 11,000 | 5,000 | 400 | 200 |
| Extra cost @end of 3rd yr | 300 | - | 100 | - | - |
| AOC | - | - | - | - | - |
| A.C for the first 5yrs | 50 | - | - | - | - |
| Cost every 6 months | - | 500 | - | - | - |
| A.I for the first 5yr | | 120 | | 120 | 75 |
| Extra cost every 5 yrs | 70 | - | 100- | - | - |
| A.I for the first 5yrswith decreasing gradient(G=40) | - | - | 300 | - | - |
| S.V | - | 1,000 | 700 | 150 | 100 |
| Life(n) yrs | ∞ | 7 | 5 | 8 | 4 |
| i% | 13% | 12%per yr compounded Quarterly | 13% | MARR=25% | MARR=25 % |

**ANSWER ANY FIVE QUESTION ONLY .**

**Q1: Using the capitalized cost criterion which is the optimal alternative A or C?**

a- CC(A) is -8675,62 8466,865 (4 marks)
b- CC(C) is -11974,52 (4 marks)
c- The decision is....A..... (2 marks)

**Q2:** a-The future worth for alternative B is..... (5 marks)
b- The AW for alternative C is ..... (5 marks)

**Q3:** Using the Rate of return(IRR) which is the optimal alternative D or E?
a- ROR(D) is ...5497,2..... (4 marks)
b- ROR(E) is ...*5314..... (4 marks)
c- The decision is....D..... (2 marks)

**Q4:a) If JD 200 is deposited each quarter for 6 years at an interest rate of 10 % yearly compounded continuously, the present worth is..... (5 marks)**

**b): The total amount due after 9 years for a loan of 9000 at 12% simple interest rate per year is..... (5 marks)**

1872

**Use i=12%**

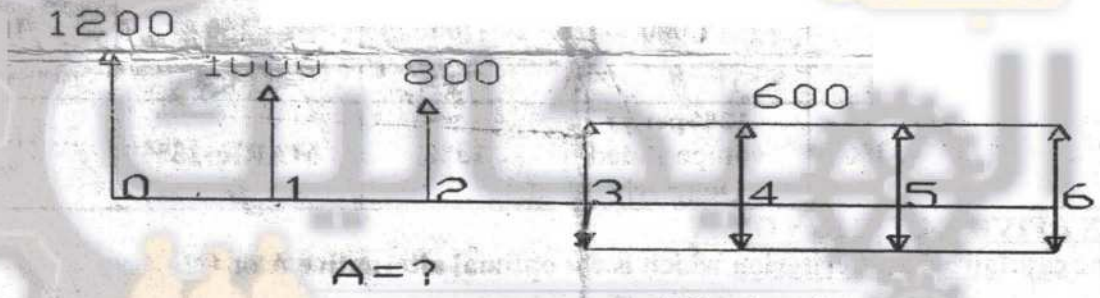
**Use the given data in the following Table to solve Q5**

| Cash Item | F | G |
|---|---|---|
| Initial (first) cost | 1800 | 2000 |
| Annual O.C | 650 | 18000 |
| Annual Income or Users income | 790 | 20000 |
| S.V | 1200 | 1100 |
| Life (years) | 6 | 3 |

**Q5:a). For Alt F, using the SOYD method**
**The depreciation at the end of year 5 is ---------- (5 marks)**

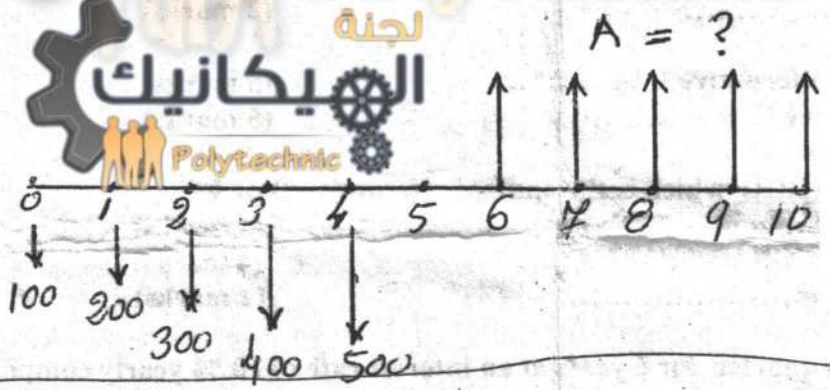
**b). For Alt. G, using SL method:**
**The book values at years 3 and 7 are:**
**$BV_2$= ----------**
**$BV_7$= ----------**
**(3 marks)**
**(2marks)**

**Q6: a). Find the Equivalent Values ( A ), (Use i.= 8% per year).---------- (5 marks)**




A=1505,76

**b) .the equation to find the equal payment amount A, in the cash flow diagram at i= 12% (5 marks)**




**a)A(P/A,12%,5)=100(F/A,12%,5) (F/P,12%,1)+100(P/G,12%,5) (F/P,12%,6)**
**b)A(P/A,12%,5)= 100(P/G,12%,6) (F/P,12%,7)**
**c)A(F/A,12%,5)=100(F/A,12%,5) (F/P,12%,6)+100(P/G,12%,5) (F/P,12%,11)**
**d) All of the above**
**e) Non of the above**